

Adversary Emulation: Build a Campaign

The design process

1. Select an actor (from a CTI report, e.g., 'APT41')
2. List their 10-ish core TTPs with ATT&CK IDs
3. Prioritize what you can emulate safely in-lab
4. Build the chain as a runbook (step → tool → detection blink)
5. Execute in a staged lab; track detection events
6. Compare against ATT&CK layer: which techniques were 'caught'?

Example mini-chain (lab-adapted)

```
Phishing (T1566) -> document macro (T1204/T1059)
-> PowerShell cradle (T1059.001)
-> beacon C2 (T1071/T1041)
-> credential dump attempt (T1003)
-> lateral AS-REP toast (T1558.004)
-> 'objective' file staged in a marked folder
```

The runbook as your deliverable

Each step:

- Technique (ATT&CK ID)
- Exact tooling + commands (repeatable)
- Expected detection (which rule *should* fire?)
- Observed result (did it? alert noise?)
- OPSEC notes

Measuring what mattered

Detection coverage map: what fired for which technique

Detection latency: when did blue's log arrive vs the action

Effectiveness: which techniques completely bypassed

The emulation report is the most actionable doc a red team produces: 'imitate actor Y, here's exactly what your detection missed, here's the rule.detection you should add'.

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